

LYNN LAKE, MB, Canada

WMO: 710780

Lat: 56.860N

Lon: 101.080W

Elev: 1170

StdP: 14.09

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.6	-32.6	-44.6	0.4	-36.3	-40.8	0.5	-32.3	21.4	-4.0	19.1	-6.7	4.8	300	0.738

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	81.4	62.2	78.4	60.6	75.1	59.1	65.2	76.3	63.2	73.5	61.4	70.7	10.1	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.5	85.2	68.0	59.5	79.1	66.0	57.5	73.8	64.9	30.8	77.0	29.3	73.5	27.9	70.6	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(4) 20.3	17.9	15.8		-43.2	87.8	4.2	2.7	-46.2	89.8	-48.6	91.4	-51.0	92.9	-54.0		
(5)				-43.2	68.8	4.1	1.9	-46.1	70.2	-48.5	71.3	-50.8	72.4	-53.7	73.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.3	-9.9	-4.7	8.3	26.3	42.1	56.2	62.0	58.6	47.5	32.3	11.4	-4.0	
	DBStd	27.70	14.18	13.34	14.77	13.37	10.07	7.89	6.08	6.45	8.28	8.35	12.79	14.94	
	HDD50	9245	1857	1531	1293	713	285	32	2	6	143	551	1157	1675	
	HDD65	13816	2322	1951	1758	1161	711	278	131	216	526	1015	1607	2140	
	CDD50	975	0	0	0	2	40	217	373	273	68	2	0	0	
	CDD65	71	0	0	0	0	1	14	37	18	1	0	0	0	
	CDH74	910	0	0	0	0	36	210	425	214	25	0	0	0	
	CDH80	158	0	0	0	0	7	33	75	39	4	0	0	0	
Wind	WSAvg	7.3	7.2	7.3	7.5	7.7	7.8	7.6	7.0	7.1	7.4	7.4	6.6	6.4	
Precipitation	PrecAvg	21.1	0.9	0.7	0.7	1.0	1.8	2.6	3.1	3.0	2.5	1.9	1.4	1.0	
	PrecMax	27.5	2.0	1.3	1.7	2.9	4.7	6.4	5.2	6.3	6.6	3.9	3.6	2.0	
	PrecMin	13.2	0.4	0.0	0.2	0.1	0.2	0.6	1.8	1.0	0.5	0.6	0.4	0.4	
	PrecStd	3.6	0.5	0.4	0.4	0.7	1.0	1.2	1.1	1.3	1.7	0.7	0.9	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.8	32.5	47.4	64.0	78.1	84.2	86.0	84.5	77.3	59.7	40.8	30.5	
		MCWB	28.7	30.4	38.5	48.7	56.4	62.6	64.6	64.6	61.4	48.9	37.0	29.1	
	2%	DB	24.8	28.2	40.7	55.7	72.2	79.6	81.9	79.6	70.2	52.6	35.2	25.3	
		MCWB	23.7	26.6	34.0	43.8	53.5	59.5	62.9	62.6	57.4	45.7	32.8	24.2	
	5%	DB	18.2	22.0	35.5	49.8	66.0	75.9	78.9	75.7	65.5	48.0	31.8	21.1	
		MCWB	17.4	20.5	31.3	40.2	50.4	58.0	61.1	60.8	55.4	43.7	30.6	20.3	
	10%	DB	11.0	16.6	30.8	45.0	60.5	71.9	75.5	71.6	60.6	44.0	28.6	17.6	
		MCWB	10.3	15.5	27.7	37.2	48.2	56.3	60.0	59.2	53.1	40.9	27.5	16.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.9	30.6	38.8	49.8	58.5	66.4	68.3	67.2	63.2	50.4	37.1	29.5
			MCDB	29.4	31.8	46.4	59.5	74.9	78.1	78.9	78.9	73.6	55.9	40.1	30.4
2%		WB	23.8	26.6	34.7	44.4	54.6	62.9	65.5	64.6	59.3	46.7	33.2	24.3	
		MCDB	24.8	28.2	39.7	54.8	67.5	72.6	76.7	76.1	67.3	51.8	34.6	25.3	
5%		WB	17.4	20.5	31.8	40.7	51.9	60.2	63.6	62.5	56.7	44.0	30.5	20.4	
		MCDB	18.2	22.0	35.6	48.6	64.7	71.3	73.9	72.6	63.9	47.7	31.9	21.0	
10%		WB	10.3	15.5	27.5	37.6	48.8	57.7	61.8	60.5	53.8	41.0	27.5	16.9	
		MCDB	10.8	16.7	30.7	44.6	59.2	69.6	71.4	68.3	60.0	43.9	28.6	17.5	
Mean Daily Temperature Range	5% DB	MDBR	16.0	19.9	22.9	22.0	21.7	20.9	19.9	18.3	16.2	11.3	13.4	14.8	
		MCDBR	20.3	21.2	24.3	25.4	30.9	27.3	26.1	25.4	24.2	18.6	13.9	14.7	
		MCWBR	19.7	19.6	18.9	15.8	16.1	11.5	10.8	11.3	13.3	12.4	11.9	14.3	
	5% WB	MCDBR	20.1	20.6	22.7	23.8	27.4	22.9	21.7	21.9	21.5	17.1	13.1	14.7	
		MCWBR	19.6	19.1	17.9	15.3	14.9	11.1	10.1	11.4	12.7	11.8	11.3	14.3	
Clear-Sky Solar Irradiance	taub	0.223	0.248	0.266	0.313	0.327	0.352	0.387	0.359	0.311	0.284	0.232	0.210		
	taud	2.233	2.279	2.287	2.303	2.467	2.441	2.368	2.464	2.573	2.559	2.358	2.158		
	Ebn at Noon	222	264	287	287	290	282	268	270	271	251	220	191		
	Edh at Noon	15	23	30	35	32	33	35	30	23	18	14	12		
All-Sky Solar Radiation	RadAvg	229	565	1084	1619	1808	1824	1692	1353	896	435	200	136		
	RadStd	22	40	77	108	165	168	124	110	114	54	19	20		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page